

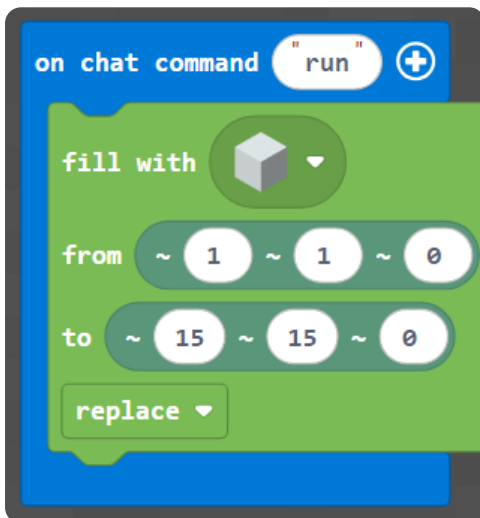
M003 Week 3 — English Worksheet (Advanced)

Topic: Pixel Art on the Wall (x, y) · **Course:** 3D Coordinates · **Level:** Advanced · **Time:** about 45 minutes

This week you **copy a picture made of blocks** onto your wall. Every block sits at a spot named by **two numbers** — (x, y). x is how far **across**, y is how far **up**. You read the picture, find each block's (x, y), and place it.

First, build your canvas. Run this to make a blank **15 × 15** wall, then put a **red block at your feet** as your **home spot**:

Blocks



Python

```
def on_run():
    blocks.fill(WHITE_CONCRETE,
                pos(1, 1, 0),
                pos(15, 15, 0),
                FillOperation.REPLACE)
    player.on_chat("run", on_run)
```

● **Big idea:** (x, y) is counted from your **home spot** — not from the corner of the world. Stand on the red block *every* time you run, or the whole picture shifts. Move your feet, move your picture.

1 · Predict 🧠

Read each set of steps. Before you imagine it happening, write what you think you will see.

```
place red block at (5, 1)
place red block at (6, 1)
place red block at (7, 1)
```

Three red blocks. Do they make a line going across or going up? Which number stays the same?


```
place green block at (8, 13)
place green block at (8, 14)
place green block at (8, 15)
```

These three share $x = 8$. Are they near the bottom of the wall or near the top? Across or up?


```
place brown block at (8, 15)
place green block at (9, 14)
place green block at (10, 14)
```

One block is brown, two are green, all near the top. Do they make part of a fruit's body, or its stem and leaf?

2 · Spot the Bug 🐛

Each block of code below was meant to do something, but it is broken. Read what the code is **supposed** to do, then rewrite it so it works. After that, explain why the original was wrong and why your fix works.

Bug A — These should make the **bottom row of the pineapple**: five yellow blocks from (6, 1) to (10, 1). Right now they all land in the **same spot**.

```
place yellow block at (6, 1)
place yellow block at (6, 1)
place yellow block at (6, 1)
place yellow block at (6, 1)
place yellow block at (6, 1)
```

Hint: to move across, the **x** has to change each time.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug B — The apple stem should go **up** at x = 8, at (8, 13), (8, 14), (8, 15). Right now it grows **across** instead.

```
place brown block at (13, 8)
place brown block at (14, 8)
place brown block at (15, 8)
```

Hint: check which slot the changing number is in — the x slot (across) or the y slot (up). The two numbers got swapped.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug C — A friend stood on the **wrong block** and ran the code. The picture came out, but it was **shifted one block too far across** — every block landed at an x one bigger than the grid says. The code is fine.

```
(the code matched the grid exactly)
```

What did your friend forget to do before pressing run? Why does standing on the home spot matter?

3 · Tell Me What You Built 📷

Now switch to your homework world and stand on your **home spot** (the red block). Pick **one** picture below and copy it onto your wall, block by block, by reading the (x, y) of every colored square. The numbers across the **top** are **x**, the numbers down the **side** are **y**.

🍷 Pineapple

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
15						■		■		■					
14						■		■		■					
13						■	■	■	■	■					
12							■	■	■						
11							■	■	■						
10					■	■	■	■	■	■	■				
9				■	■	■	■	■	■	■	■	■			
8				■	■	■	■	■	■	■	■	■			
7			■	■	■	■	■	■	■	■	■	■	■	■	
6			■	■	■	■	■	■	■	■	■	■	■	■	
5			■	■	■	■	■	■	■	■	■	■	■	■	
4			■	■	■	■	■	■	■	■	■	■	■	■	
3			■	■	■	■	■	■	■	■	■	■	■	■	
2				■	■	■	■	■	■	■	■				
1					■	■	■	■	■	■					



	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
15															
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1															

When your picture is finished, send a photo of it, then explain what you did. Use these sentence starters — write 4 to 6 sentences total.

First, I chose to build the ...

The lowest block in my picture is at $(x = \dots, y = \dots)$.

To go across a row I changed the ... number; to go up I changed the ... number.

The trickiest part to read off the grid was ...

One time my picture landed in the wrong place because ...

If I had more time, I would ...

4 · Record Your Walkthrough

Now take a video on your phone (or a parent's phone) while you show your pixel art on the wall. Talk through it like you are teaching someone who has never seen it. Try to use these words: **x, y, coordinate, across, up, home spot.**

1. Show your home spot (the red block) and say why you stand on it every time.
2. Show your finished picture from the front.
3. Point at one block and read its coordinate out loud: "this one is at x ..., y ...".
4. Show one block that you placed in the wrong spot at first, and how you fixed it.

Write what you will say in your video. Use the space below to plan it before you record — you can read from it while filming.

Submit 

Send this worksheet + your walkthrough video to teacher on KakaoTalk.